Linked List - 4

Agenda:

1. Doubly Linked List Intro
2. Insert a node in DLL
3. Delete the first occurence of an element in DLL
4. K Most Recent Integer (LRU Cache)

① Doubly Linked List

```
class Node {
    int data;
    Node prev;
    Node next;

    Node (int x) {
        this.data = x
        this.prev = null;
        this.next = null;
    }
}
```

- **Prev:** Points to the node before it (the first node's prev is always null).
- **next:** Points to the node after it (the last node's next is always null)

② Inserting a Node at Index K

```
public Node insertAtK (Node head, int x, int k){
    Node newNode = new Node(x);
    if (k==0) {
        newNode.next = head;
        if (head != null) head.prev = newNode.
        return newNode;
    }

    Node temp = head;
    for (int i=0; i< k-1 && temp != null ; i++){
        temp = temp.next;
    }
    if (temp != null){
        newNode.next = temp.next;
        newNode.next = temp;
        if (temp.next != null) temp.next.prev = newNode.
        temp.next = newNode;
    }
    return head;
}
```

T.C: O(k)
because you have to walk to the kth position

③ **Deleting the first Occurence of X**

```
public Node deleteNode (Node head, int x) {
    Node temp = head;
    while (temp != null) {
        if (temp.data == x) break;
        temp = temp.next;
    }
    if (temp == null) return head;
    if (temp.prev != null) temp.prev.next = temp.next;
    else head = temp.next;
    if (temp.next != null) temp.next.prev = temp.prev;
    return head;
}
```